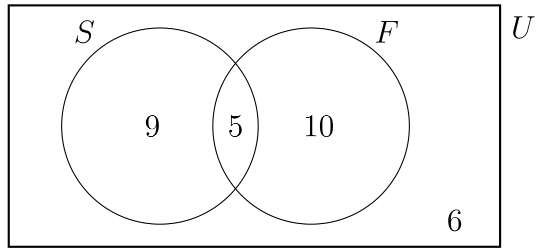
4.18 Homework: Probability Review — Solutions

1. In a class of 30 students, some study Spanish (S) and some study French (F). Five students study both languages.



   (a) Write down the number of students who study Spanish. [1]

   (b) One student is chosen at random. Find the probability that

      (i) the student does not study Spanish; [1]

      (ii) the student studies Spanish or French, but not both. [2]

> **Solutions**
>
> **(a)** $n(S) = 9 + 5 = \mathbf{14}$
>
> **(b)(i)** $\mathrm{P}(S') = \dfrac{10+6}{30} = \dfrac{16}{30} = \dfrac{\mathbf{8}}{\mathbf{15}}$
>
> **(b)(ii)** Spanish or French but not both $= 9 + 10 = 19$
>
> $\mathrm{P} = \dfrac{19}{30}$

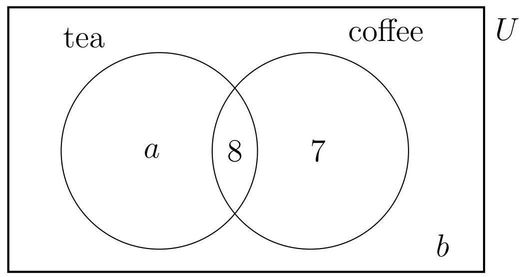
2. Events C and D are mutually exclusive with $\mathrm{P}(C) = 0.35$ and $\mathrm{P}(D) = 0.25$.

   (a) Write down $\mathrm{P}(C \cap D)$. [1]

   (b) Find $\mathrm{P}(C \cup D)$. [2]

> **Solutions** Check formula sheet for *mutually exclusive*
>
> **(a)** $\mathrm{P}(C \cap D) = \mathbf{0}$
>
> **(b)** $\mathrm{P}(C \cup D) = \mathrm{P}(C) + \mathrm{P}(D) - \mathrm{P}(C \cap D)$
>
> $= 0.35 + 0.25 - 0 = \mathbf{0.6}$

3. In a group of 40 people, 22 like tea and 15 like coffee. Eight people like both, as shown in the Venn diagram. The values $\boldsymbol{a}$ and $\boldsymbol{b}$ represent numbers of people.



(a) Find the value of a. [1]

(b) Find the value of b. [1]

(c) A person is chosen at random. Find the probability that the person likes coffee but not tea. [1]

(d) Find the probability that a person likes tea given that they like coffee. [2]

Solutions

(a) $a + 8 = 22 \;\Rightarrow\; a = \mathbf{14}$

(b) $14 + 8 + 7 + b = 40 \;\Rightarrow\; b = \mathbf{11}$

(c) Coffee but not tea is the "coffee only" region $= 7$

$$\mathrm{P}(\text{coffee} \cap \text{tea}') = \frac{7}{40}$$

(d) $\mathrm{P}(\text{tea} \mid \text{coffee}) = \dfrac{\mathrm{P}(\text{tea} \cap \text{coffee})}{\mathrm{P}(\text{coffee})} = \dfrac{8/40}{15/40} = \dfrac{8}{15}$

4. The following table shows the travel method and punctuality of 20 students.

	Punctual (P)	Sometimes late (T)	Total
Walk	3	7	10
Bus	2	8	10
Total	5	15	20

(a) Find the probability that a randomly chosen student walks to school and is punctual. [1]

(b) Find P(punctual | bus). [2]

(c) Determine whether "walks to school" and "is punctual" are independent events. Show your working and justify your answer. [3]

Solutions

(a) $\mathrm{P}(W \cap P) = \dfrac{3}{20}$

(b) $\mathrm{P}(P \mid B) = \dfrac{\mathrm{P}(P \cap B)}{\mathrm{P}(B)} = \dfrac{2/20}{10/20} = \dfrac{2}{10}$

(c) If independent, then $\mathrm{P}(W \cap P) = \mathrm{P}(W) \times \mathrm{P}(P)$

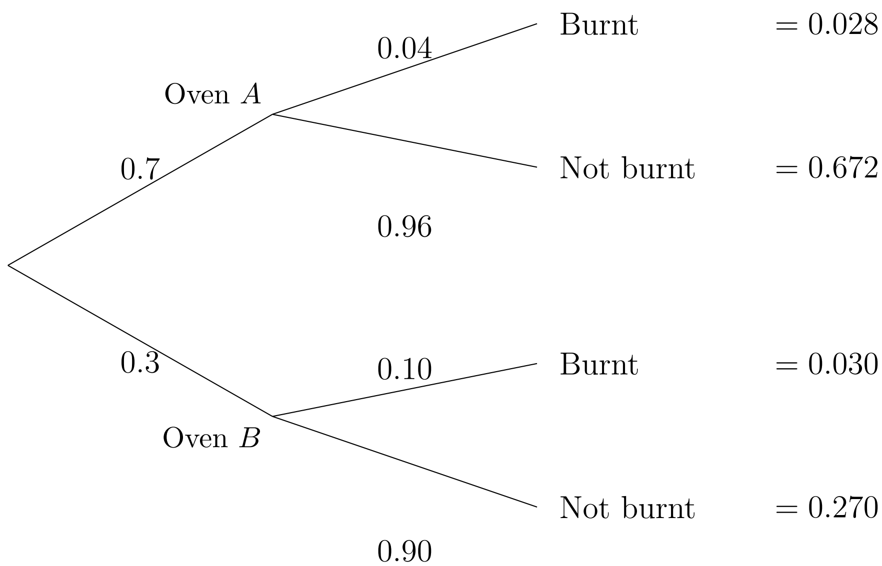
$\mathrm{P}(W) \times \mathrm{P}(P) = 0.5 \times 0.25 = 0.125$

$\mathrm{P}(W \cap P) = \dfrac{3}{20} = 0.15$

$0.15 \neq 0.125$, so the events are **not independent**.

5. A bakery makes bread using two ovens. Oven A bakes 70% of the loaves and oven B bakes the remaining 30%. The probability that a loaf from oven A is burnt is 0.04. The probability that a loaf from oven B is burnt is 0.10.

(a) Complete the tree diagram below, showing all probabilities. [3]



(b) A loaf is selected at random. Find the probability that the loaf is

(i) burnt and from oven A; [2]

(ii) burnt. [2]

(c) Given that a loaf is burnt, find the probability that it came from oven A. [3]

Solutions

(a) Blanks: $1 - 0.04 = 0.96$; 0.10; $1 - 0.10 = 0.90$

Composite: $0.7 \times 0.04 = 0.028$; $0.7 \times 0.96 = 0.672$

$0.3 \times 0.10 = 0.030$; $0.3 \times 0.90 = 0.270$

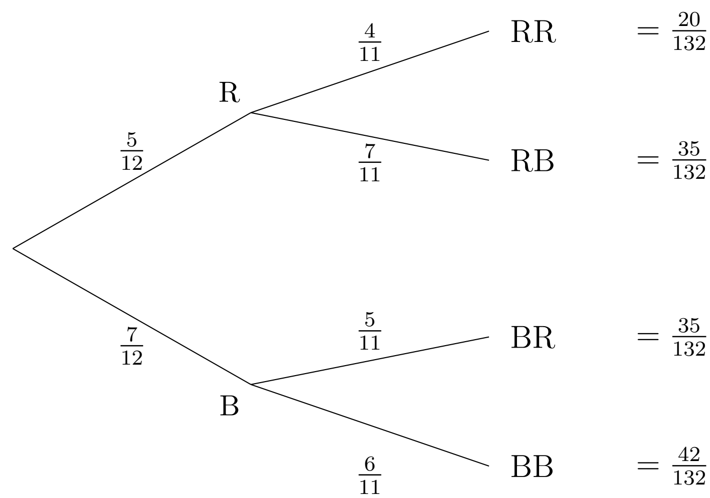
(b)(i) $\mathrm{P}(A \cap \mathrm{Burnt}) = 0.7 \times 0.04 = \mathbf{0.028}$

(b)(ii) $\mathrm{P}(\mathrm{Burnt}) = 0.028 + 0.030 = \mathbf{0.058}$

(c) $\mathrm{P}(A \mid \mathrm{Burnt}) = \dfrac{\mathrm{P}(A \cap \mathrm{Burnt})}{\mathrm{P}(\mathrm{Burnt})} = \dfrac{0.028}{0.058} = 0.48275\ldots \approx \mathbf{0.483}$

6. A bag contains 5 red marbles and 7 blue marbles. Two marbles are drawn at random, one after the other, without replacement.
   (a) Find the probability that both marbles are red. [2]
   (b) Find the probability that both marbles are the same colour. [3]
   (c) Find the probability that both marbles are red given that at least one marble is red. [3]

A tree is a good way to organize these two-step problems.



Solutions

(a) $\mathrm{P}(RR) = \dfrac{5}{12} \times \dfrac{4}{11} = \dfrac{20}{132} = \dfrac{5}{33}$

(b) $\mathrm{P}(BB) = \dfrac{7}{12} \times \dfrac{6}{11} = \dfrac{42}{132} = \dfrac{7}{22}$

$$\mathrm{P}(\text{same}) = \mathrm{P}(RR) + \mathrm{P}(BB) = \frac{20}{132} + \frac{42}{132} = \frac{62}{132} = \frac{31}{66}$$

(c) $\mathrm{P}(RR \mid \text{at least one } R) = \dfrac{\mathrm{P}(RR)}{\mathrm{P}(\text{at least one } R)}$

$$\mathrm{P}(\text{at least one } R) = 1 - \mathrm{P}(BB) = 1 - \frac{42}{132} = \frac{90}{132}$$

$$\mathrm{P}(RR \mid \text{at least one } R) = \frac{20/132}{90/132} = \frac{20}{90} = \frac{2}{9}$$